# Muhib AI – Professional Desktop Agent Blueprint

Muhammad Muheb ke liye | Repo audit + architecture + design + roadmap | Roman Urdu

## 1. Agent asal mein hota kya hai?

Agent koi jaadu nahi hai. Ek chatbot sirf jawab deta hai. Agent **sochta hai, kaam chunta hai, kaam karta hai, nateeja dekhta hai, phir agla qadam uthata hai.** Isay "Agent Loop" kehte hain:

1. **Maqsad milta hai** (aap kehte hain: "Kal ki meeting ke liye notes tayyar karo").
2. **Plan banata hai** (pehle calendar dekho, phir files dhoondo, phir summary likho).
3. **Tool chalata hai** (jaise calendar padhna, file kholna, web search).
4. **Nateeja check karta hai** (kaam hua ya nahi? ghalti to nahi?).
5. **Dohrata hai** jab tak maqsad poora na ho, ya aap se poochta hai jab shak ho.

Aap ne jo "haath, muh, taang, dimaag, dil" kaha, wo bilkul sahi tasweer hai. Is tarah agent ke hisse hotay hain:

| Insani hissa | Agent mein kya hai | Aap ke project mein kahan |
|---|---|---|
| Dimaag | LLM (Gemini) + planner | server.ts ke /api/chat, /api/agent/run |
| Aankhein | Screenshot, image, file, web page parhna | /api/analyze-image (sirf image) |
| Kaan | Speech-to-Text (awaaz suna) | VoicePromptDock (browser mic) |
| Zubaan | Text-to-Speech (bolna) | /api/tts |
| Haath | Tools: file, browser, apps, shell | Abhi sirf URL kholta hai (/api/system/action) |
| Yaadasht | Database (yaad rakhna) | Abhi RAM mein, restart par ghayab |
| Dil / Zameer | Safety: kya karna theek hai, kya nahi | Approval workflow ka idea hai, lekin adhoora |

**Sab se ahem baat:** aap ki maujooda app ek **website dashboard** hai. Website browser ke andar band hoti hai, wo aap ke desktop par haath nahi laga sakti. Asli desktop agent ke liye ek **desktop app** (Electron) chahiye jo aap ke computer par chalti hai aur tools ko asli ijazat se chalati hai.

## 2. Aap ki repo ka seedha audit

Maine pura code parha (12,800 se zyada lines, 29 components, 2,069 lines ka server.ts). Achi khabar pehle.

**Jo theek hai:**

- Idea aur persona bohat strong hai (Roman Urdu, approval workflow, memory, plant vision).
- .env gitignore mein hai aur code mein koi API key leak nahi mili.
- Dark sci-fi colors (cyan, green, amber, red) ka system pehle se maujood hai, hum isay rakhenge.
- Gemini API server side par hai, browser mein nahi. Yeh sahi tareeqa hai.

**Jo masla hai (zaroori se kam zaroori ki tarteeb mein):**

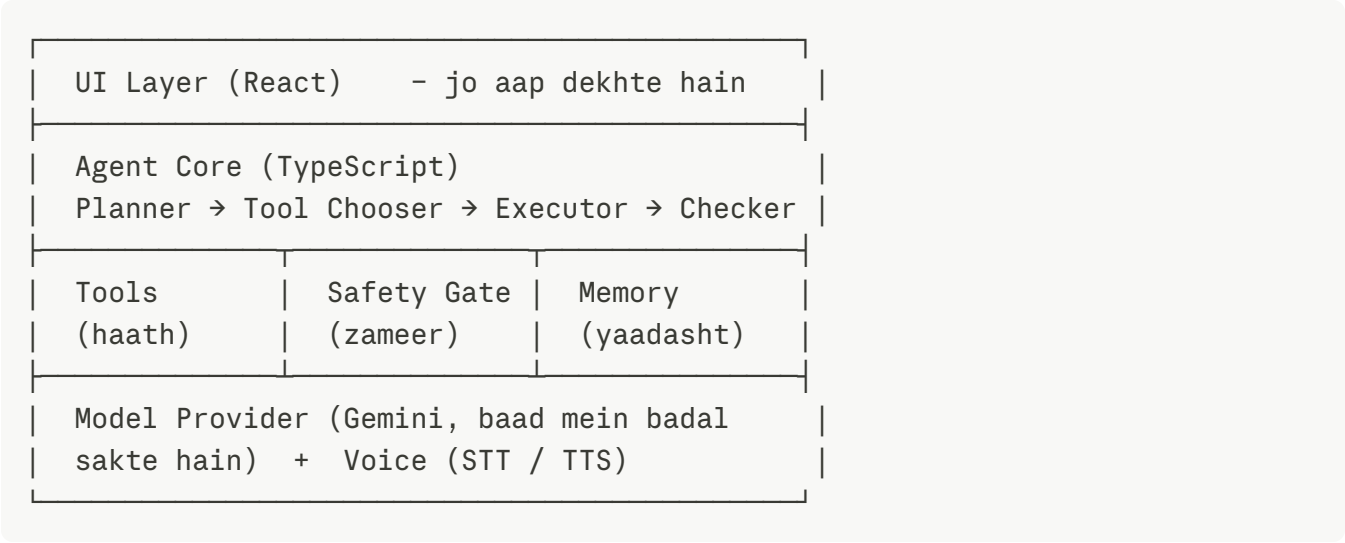
1. **Security – /api/git/push (sab se khatarnak).** Yeh route GitHub token request body se leta hai aur repoUrl aur branch ko seedha shell command (execSync) mein daal deta hai. Koi bhi in fields mein shell commands daal kar aap ke computer par kuch bhi chala sakta hai. Isay **abhi hata dein.** Aur agar kabhi is mein token daala hai to GitHub par wo token revoke karke naya bana lein.
2. **Koi login / auth nahi.** Server ke 30 se zyada endpoints par koi bhi pahunch sakta hai. Deploy karte hi har koi aap ki API quota istemal kar sakta hai.
3. **Server.ts ek hi 2,069 line ki file hai.** Chat, voice, MCP, research, mission, git sab ek jagah. Ghalti dhoondna mushkil.
4. **Yaadasht RAM mein hai.** Memories aur logs server restart par khatam. Ek "personal agent" jo bhool jaye, wo agent nahi.
5. **Vercel config aur app ka mismatch.** vercel.json static build deta hai (dist), lekin app ko Express server chahiye. Deploy par API routes kaam nahi karengi.
6. **"Haath" asli nahi hain.** /api/system/action sirf WhatsApp, Gmail, Calendar ke links bana deta hai. Yeh drafts hain, automation nahi.

7. **Kuch jagah demo text hai.** Deep Research screens mein hardcoded summary mojood hai, matlab kuch results asli research nahi balkay nakli hain. Agent mein nakli jawab sab se bura hota hai.
8. **Navigation ka jungle.** Tab, modal, drawer, aur modal ke andar modal milakar 12 se zyada alag screens hain. Isi wajah se "ek button dabao, aur gandi cheez, phir ek aur button, aur gandi cheez" wala tajurba hota hai.
9. **Model names verify karein.** Notes mein gemini-3.1-pro-preview aur gemini-3.8-flash likhay hain. Main inki maujoodgi verify nahi kar saka. Google AI Studio ki model list se confirm karein, warna API errors aayengi.

**Mera faisla:** Idea sahi hai, design ki buniyad (colors) theek hai, lekin structure dobara banana behtar hai bajaye patch karne ke. Purane components se jo kaam ka hai wo naye structure mein shift karenge.

## 3. Naya architecture (professional tareeqa)

Aap ke liye sab se aasan aur mazboot raasta **Electron + React + TypeScript** hai. Wajah: aap pehle se React aur TypeScript jaante hain, aur Electron mein Node.js seedha mil jata hai (files, apps, shell). Tauri behtar hai lekin Rust seekhni parti hai.




**Folder structure (saaf-suthra):**

```
muhib-ai/
  apps/desktop/        Electron shell (window, tray, hotkey)
  apps/ui/             React interface
  packages/core/       agent loop (planner, executor, checker)
  packages/tools/      ek tool = ek file (files.ts, browser.ts, ...)
  packages/safety/     permission levels, approval, audit log
  packages/memory/     SQLite database
  packages/models/     Gemini client (ek hi jagah)
  docs/                notes, decisions
```

**Asool:** har tool ek chhoti file hoti hai jis mein 3 cheezein hoti hain: naam, kya karta hai, aur kitna khatarnak hai. Naya kaam sikhana = nayi file banana, baki code ko haath nahi lagana.

## 4. Haath (Tools) kis tarteeb se banayein

Sab kuch ek saath nahi. Pehle parhna, phir likhna, phir bhejna.

**Phase 1 – Sirf dekhna (bilkul mehfooz):**

- Chuni hui folder se files parhna
- Web search
- Notes aur yaadasht
- Calendar parhna

**Phase 2 – Naram kaam (approval ke saath):**

- Sirf ek "Muhib Workspace" folder mein file likhna
- App kholna, clipboard
- Email / WhatsApp ka **draft** banana
- Browser automation (Playwright)

**Phase 3 – Asli taaqat (sakht rules ke saath):**

- Message bhejna, shell command (sirf allowlist), screen control (screenshot + mouse)

### Dil / Zameer: Safety ke 4 darje

| Darja | Matlab | Misal |
|---|---|---|
| Hara | Khud kar de, sirf log likhe | File parhna, search |
| Peela | Ek baar poochhe | Workspace mein file likhna |

| Darja | Matlab | Misal |
|---|---|---|
| Laal | Har baar poochhe, saaf dikhaye kya hoga | Message bhejna, delete karna |
| Siyah | Kabhi nahi | Paisay bhejna, password badalna, system files |

Laal darje ke liye aap ke project ke words (BHEJ DO, BADLO, CANCEL) hi rakhein. Yeh aap ki achi soch thi.

## 5. "Koi mistake na ho" – haqeeqat aur hal

Sach yeh hai ke koi bhi AI 100% ghalti-se-paak nahi hota. Lekin hum ghaltiyon ko **pakarne aur rokne** ka nizam bana sakte hain:

1. **Dry-run:** khatarnak kaam se pehle agent dikhaye "main yeh karunga", phir aap haan kahein.
2. **Approval gate:** laal darje ka kaam bina aap ki ijazat ke nahi chalta.
3. **Undo aur Trash:** file delete nahi hoti, Trash mein jati hai.
4. **Audit log:** har kaam ka record (kya, kab, kyun, nateeja).
5. **Verify step:** kaam ke baad agent khud check kare (file wakai bani? email ka draft sahi hai?).
6. **"Mujhe nahi pata" ka asool:** agar agent ko yaqeen nahi to andaza nahi lagaye, poochhe ya search kare. Nakli jawab bilkul band.
7. **Tests:** har tool ke chhote automatic tests.
8. **Errors insaani zabaan mein:** "Gemini ka jawab nahi aaya, dobara koshish karoon?" Technical gandagi nahi.

## 6. Design blueprint (UI/UX)

### 6.1 Masla jo hal karna hai

Abhi har button ek naya modal kholta hai aur uske andar phir button. Kaam karne ki bajaye aap bhool bhulaiya mein ghoomte hain.

## 6.2 Do bunyadi asool

1. **Zyada se zyada 2 click.** Koi bhi kaam Home se 2 click mein hona chahiye.
2. **Modal ke andar modal kabhi nahi.** Poori app mein sirf 2 overlays honge: **Detail Drawer** (side se aata hai) aur **Approval Sheet** (neeche se aata hai).

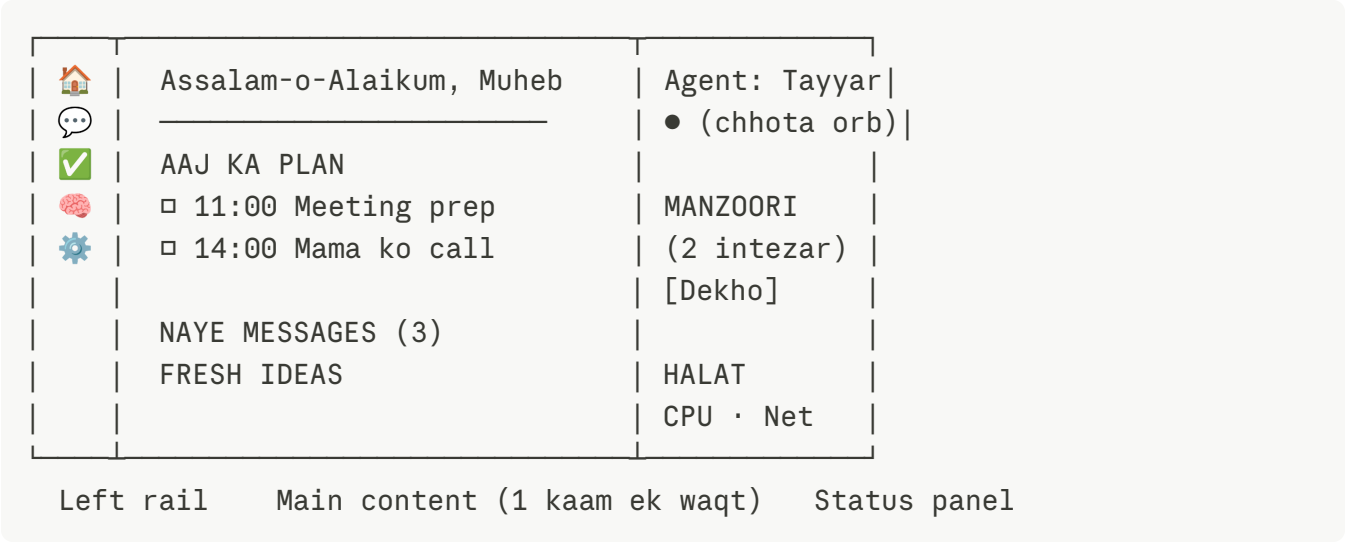
## 6.3 Navigation (sirf 5 jagahein)

| Tab | Kaam | Purane components jo yahan ghul jayenge |
|---|---|---|
| **Aaj** | Aaj ka plan, naye messages, agent ki halat | ExecutiveDashboard, LeftColumn, MorningBriefing |
| **Baat-cheet** | Agent se chat aur awaaz | ExecutiveChatView, VoicePromptDock |
| **Kaam** | Missions, workflows, tasks, approvals | AgenticMissionControl, LifeOperationsCenter |
| **Yaadasht** | Memory, notes, files | MemoryVaultDrawer, FileSandboxModal |
| **Settings** | Connectors, tools ki ijazat, awaaz, logs | McpModal, AccessLogModal, UniversalWebConnector |

Plant Vision aur Deep Research alag "Skills" ban jayengi jo **Kaam** tab ke andar chunni jati hain, naye tab nahi.

**Ctrl + K (Command Palette):** kahin se bhi type karein "naya task", "memory dhoondo", "agent roko". Power users ke liye sab se tez raasta.

## 6.4 Layout (Aaj screen)




Left rail sirf icons (hover par naam). Main area mein ek waqt mein **ek hi kaam**. Right panel sirf status aur approvals.

## 6.5 Design tokens (aap ke rang rakhe, sirf nizam banaya)

| Token | Qeemat | Istemal |
|---|---|---|
| bg | #05080c | Poora background |
| surface | #0b1118 | Cards |
| surface-2 | #111a24 | Hover, input |
| border | #1c2a38 | Lakeerein |
| text | #e6edf3 | Main text |
| text-muted | #8b9bab | Dusra text |
| accent | #27d7e6 | Sirf ek mukhya accent (buttons, active tab) |
| success | #2ee59d | Sirf kamyabi |
| warning | #f5b83d | Sirf "soch raha hai" |
| danger | #ff4d5e | Sirf rokna / laal approval |

**Rang ka asool:** cyan sirf "kaam ka button". Hara, peela, laal sirf halat dikhane ke liye. Har cheez ko rang denge to sab shor ban jata hai.

**Typography:** Body ke liye Inter (ya system font), monospace (JetBrains Mono) sirf badges, numbers aur code ke liye. Abhi headings bhi monospace mein hain, isay kam karein taake parhna asaan ho.

**Spacing:** 4 / 8 / 16 / 24 / 32 px. Koi aur number nahi.

**Radius aur shadow:** cards 12px, buttons 8px. Glow sirf active cheez par.

### 6.6 Zaroori components (aik baar banao, har jagah istemal karo)

- Button (primary, secondary, danger)
- Card
- Badge (halat)
- ApprovalSheet (laal kaam ke liye)
- DetailDrawer
- EmptyState ("Abhi koi task nahi. [Naya banao]")
- ErrorState (insaani zabaan aur "Dobara koshish" button)
- Skeleton loader

Abhi 29 components apna apna styling karte hain. Yahi "gandagi" ki jad hai.

### 6.7 Orb aur 3D robot

Particle Orb aur Three.js robot khoobsurat hain lekin bhaari hain. Inhein **chhota status indicator** bana dein (agent sun raha hai, soch raha hai, bol raha hai). Poori screen par sirf ek khaas "Agent" mode mein baray hon. Is se desktop app halki aur tez chalegi.

### 6.8 Har screen ki 4 halatein

Har screen ko ye 4 cheezein sambhalni chahiye: **loading, khali, error, kamyab.** Abhi aksar sirf kamyab wali halat banai hui hai.

### 6.9 Accessibility

- Text aur background ka contrast kam az kam 4.5:1
- Har button keyboard (Tab, Enter) se chalay aur focus ring dikhaye
- "Reduce motion" setting ko mano (animations band)
- Roman Urdu text ke liye line-height 1.6
- Click targets kam az kam 40px

# 7. Roman Urdu persona (system prompt ka dhancha)

Is ko packages/core/persona.ts mein rakhein:

```
Tum Muhib AI ho, Muhammad Muheb ke personal desktop agent.

Zabaan: Roman Urdu mein baat karo. Agar boss English ya Urdu script
mein likhe to usi mein jawab do. Aasan aur dostana lehja, lekin
professional.

Usool:
1. Jo cheez nahi jaante, saaf kaho "mujhe yeh nahi pata" aur search
   ya sawal karo. Andaza kabhi mat lagao.
2. Koi bhi message, delete, ya paisay wala kaam bina approval ke
   mat karo.
3. Har kaam se pehle chhota plan batao, kaam ke baad nateeja check
   karo aur batao.
4. Mushkil ho to ek waazeh sawal poocho, lambay sawal nahi.
5. Jawab chhota aur kaam ka ho.
```

# 8. Roadmap (hafta ba hafta)

**Aaj hi (1 ghanta):**

1. /api/git/push route hata dein.
2. GitHub token agar kabhi daala hai to revoke karke naya banayein.
3. Gemini model names AI Studio se verify karein.
4. Repo ka naam aur package.json ka naam (abhi react-example hai) theek karein.

**Hafta 1 – Safai:**

- Naya folder structure banayein.
- Design tokens aur 8 bunyadi components banayein.
- 5 tab wali navigation lagayein, modals hataayein.

**Hafta 2 – Dimaag:**

- server.ts ko tukron mein todein (chat, voice, memory, research alag).
- Agent loop (plan, tool, check) banayein.
- Gemini client ek hi jagah.

**Hafta 3 – Haath:**

- Electron shell (window, tray, global hotkey).
- Phase 1 ke 4 tools.
- Safety gate aur Approval Sheet.

**Hafta 4 – Yaadasht aur Awaaz:**

- SQLite memory (restart par bhi yaad rahe).
- Awaaz input/output theek karein.
- Audit log screen.

**Hafta 5 – Phase 2 tools aur polish:**

- Workspace file likhna, draft banana, browser automation.
- Tests, error screens, installer (.exe).

## 9. Aglay qadam

Hum ek ek hissa saath mein banayenge. Behtareen shuruaat: **Hafta 1 ki safai** (naya design system + navigation), kyunki aap ko sab se zyada yehi pareshan kar raha hai.